

# Changing Dimension in FFGFT

Geometric Reduction ( $T^4 \rightarrow T^0$ ) and Representational Translation

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## Contents

|     |                                                                                              |    |
|-----|----------------------------------------------------------------------------------------------|----|
| .1  | Motivation: the tension in the question . . . . .                                            | 2  |
| .2  | Structure of $T^4$ . . . . .                                                                 | 3  |
|     | Decomposition and radius classes . . . . .                                                   | 3  |
|     | Kaluza–Klein reading . . . . .                                                               | 3  |
| .3  | Two types of reduction . . . . .                                                             | 4  |
|     | Type I: duality decompactification (mass $\rightarrow$ time) . . . . .                       | 4  |
|     | What makes the unrolled time continuous and uniform . . . . .                                | 4  |
|     | Type II: geometric projection (spatial) . . . . .                                            | 5  |
| .4  | The formal chain, step by step . . . . .                                                     | 6  |
|     | Step 1: $D_4 \rightarrow D_3 + \text{time}$ (Type I) . . . . .                               | 6  |
|     | Step 2: $D_3 \rightarrow D_2$ (Type II, holographic) . . . . .                               | 6  |
|     | Step 3: $D_2 \rightarrow D_1$ (Type II) . . . . .                                            | 6  |
|     | Step 4: $D_1 \rightarrow D_0$ (Type II, total loss) . . . . .                                | 6  |
| .5  | Reversibility, precise per type . . . . .                                                    | 6  |
| .6  | Role of the recursion $\xi^n$ . . . . .                                                      | 7  |
| .7  | Overview . . . . .                                                                           | 7  |
| .8  | The upward direction: representational translation (Type III) . . . . .                      | 7  |
| .9  | Why exactly four circles? Minimal sufficiency, units, and visualization . . . . .            | 8  |
|     | $3 + 1$ is input, not derivation . . . . .                                                   | 8  |
|     | Minimal sufficiency: four instead of five . . . . .                                          | 9  |
|     | The four circles as units bookkeeping . . . . .                                              | 9  |
|     | The same structure as Hilbert visualization . . . . .                                        | 9  |
|     | Summary of the dimension line . . . . .                                                      | 10 |
| .10 | In the compact view, time is just another circle: the asymmetry is the measurement . . . . . | 10 |
| .11 | Physical testability of the identification $t \sim t + R_m$ . . . . .                        | 11 |
|     | Two testable readings . . . . .                                                              | 11 |
|     | Scaling: linear versus logarithmic . . . . .                                                 | 11 |
| .12 | Classification and open questions . . . . .                                                  | 12 |

### Correction notes

**Corrections from the calculation review (1 Oct. 2026).** The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Abstract:  $t \sim t + R_m$  "shown to be" an observable  $\rightarrow$  treated as one; CHSH deviation  $\sim \xi$  not resolvable with present hardware (Doc. 190 R142/R143).
- Time reading and conclusion: "currently just below resolution", "resolvable at  $\sigma \sim 10^{-4}$ "  $\rightarrow$  far below the hardware noise ( $2\sqrt{2} - 2.74 = 0.088$ , about  $160\Delta$ ), not resolvable with present hardware.
- (2 Oct. 2026) Notation  $f$ : correction function in  $g = \eta(1 + \xi f)$  — not a frequency; note in the text (Doc. 190, R147).

## Abstract

The reduction of the fundamental 4-torus structure  $T^4$  of FFGFT to lower dimensions is examined as a formal chain  $T^4 \rightarrow T^3 \rightarrow T^2 \rightarrow T^1 \rightarrow T^0$ . The central claim is that this chain does *not* consist of steps of the same kind. It splits into two qualitatively distinct operations. The first step ( $D4 \rightarrow D3$ ) is not a lossy projection but a *duality decompactification*: the compact mass dimension  $S_m^1$  is unrolled into the non-compact time direction via the fundamental relation  $\tilde{T} \cdot m = 1$ . This operation preserves the mode content; only the direct visibility of compactness survives, as a discrete spectral signature encoded by the recursion  $\xi^n$ . The further steps ( $D3 \rightarrow D2 \rightarrow D1 \rightarrow D0$ ) are genuine geometric projections of spatial directions: strictly lossy, not reversible, and *not* compensated by the recursion. The step  $D3 \rightarrow D2$  is identified as a holographic projection. The topological identification  $t \sim t + R_m$  is treated as a physical observable ( $T_{\text{rev}}$ , CHSH deviation  $\sim \xi$ , the latter not resolvable with present hardware). (*Corrected on 1 Oct. 2026: previously "shown to be a physical observable" without qualification – by Doc. 190 R142/R143 the  $\xi$  effect in CHSH is not resolvable with present hardware; the only documented hardware measurement (Doc. 147) confirms only the Bell violation.*) The downward reduction is contrasted with the *upward* operation: the bijective representational translation  $T^4 \leftrightarrow$  Hilbert space/matrix (Type III, Doc. 230/206/231), which is *lossless* – precisely because it adds no geometric extra dimensions but expresses the same 4D content in a higher-dimensional language. The work thereby resolves the apparent tension between "projection loses information" and "compactified / decompactified", and classifies geometric downward reduction against representational upward translation.

## 1 Motivation: the tension in the question

FFGFT describes fundamental physics on a compact 4-torus  $T^4$  with the single dimensionless parameter  $\xi = 4/30000$  and the central relation  $\tilde{T} \cdot m = 1$ . The observable world, by contrast, is a three-dimensional space with time evolution,  $\mathbb{R}^3 \times \mathbb{R}_t$ .

Three observations on this transition appear to conflict:

- Loss picture.** Every projection  $D \rightarrow D-1$  is not injective and therefore loses information. The chain  $T^4 \rightarrow \dots \rightarrow T^0$  is not reversible.
- Compactness picture.**  $T^4$  is fully compactified; the observable spacetime  $\mathbb{R}^3 \times \mathbb{R}_t$  is decompactified.
- Upward picture.** There already exists a *bijective* translation of  $T^4$  into a higher-dimensional space – the Hilbert space (Doc. 230) – which is *lossless*.

These three pictures cannot be naively superimposed: if the first step were pure loss (A), no *decompactified* time direction (B) could arise from it, since forgetting creates no new dimension; and it would remain puzzling how an upward move (C) can be lossless while the downward move (A) destroys information.

The resolution: there is not one but *three* dimension-changing operations of *two kinds*. *Geometric* operations (Type I: mass-time duality; Type II: spatial projection) reduce the observable world out of  $T^4$ ; the *representational* operation (Type III) translates  $T^4$  upward without loss. The geometric-downward lossy versus representational-upward bijective move – this asymmetry is the central result of the work.

## .2 Structure of $T^4$

### Decomposition and radius classes

**Definition .2.1** (Fundamental torus structure). The fundamental level of FFGFT is the compact 4-torus

$$T^4 = S_x^1 \times S_y^1 \times S_z^1 \times S_m^1,$$

with three *spatial* circles  $S_{x,y,z}^1$  and one *mass circle*  $S_m^1$ . All four components are compact.

*Remark .2.2* (The fourth direction as substitute time). The fourth torus direction  $S_m^1$  is not a mass dimension *alongside* a separate time, but the *substitute-time direction* itself. Mass and time are two *readings* of one and the same dimension; the fundamental relation  $\tilde{T} \cdot m = 1$  is the translation dictionary:

$$\underbrace{S_m^1 \text{ (compact)}}_{\text{mass reading, discrete spectrum}} \xleftrightarrow{\tilde{T} \cdot m = 1} \underbrace{R_t \text{ (open)}}_{\text{time reading, continuous flow}}.$$

There is no time “in addition” to the four directions; time is the decompact reading of the fourth direction (cf. Doc. 185). The two-type structure depends only on the separation of *small* and *large* compactification radii and is independent of the labelling.

The four directions carry two separated radius classes:

$$R_x = R_y = R_z \sim R_H \quad (\text{cosmological horizon, “large”}), \quad (1)$$

$$R_m \sim \xi \cdot \ell_P \quad (\text{Planck scale, “small”}). \quad (2)$$

*Remark .2.3* (Status of the two scales).  $R_H$  here is the *declared external* (imported) cosmological scale, *not* derived from  $\xi$ . Intrinsic is only the *form* – the large/small radius-class split, equivalently the dimensionless ratio  $R_H/\ell_P$  – not the absolute *factor* (Doc. 190, P20). The two-type structure of this document depends only on that class split, not on the numerical value of  $R_H$ . Conversely  $R_m \sim \xi \ell_P$  sets the *top* of the KK tower ( $1/R_m \sim \xi^{-1} m_P$ , trans-Planckian); the observed particle masses lie further down the ladder via the recursion  $\xi^n$  (Doc. 159, 258/259).

### Kaluza–Klein reading

A compact circle of radius  $R$  carries a discrete mass tower  $m_n \sim n/R$ ,  $n \in \mathbb{Z}$ . The two radius classes therefore generate qualitatively different physics:

| Circle        | Radius                      | Mode tower                                                        |
|---------------|-----------------------------|-------------------------------------------------------------------|
| $S_{x,y,z}^1$ | $R \sim R_H$ (large)        | gap $\sim 1/R_H \approx 0$ , quasi-continuous $\Rightarrow$ space |
| $S_m^1$       | $R \sim \xi \ell_P$ (small) | gap huge, discrete and heavy $\Rightarrow$ mass spectrum          |

**Proposition .2.4** ( $\tilde{T}m = 1$  as compactification relation). *In natural units a compactification radius  $R$  has the dimension of a length or intrinsic time,  $\tilde{T} \sim R$ , while the generated mass scale is  $m \sim 1/R$ . The fundamental relation*

$$\tilde{T} \cdot m = 1$$

*is thus structurally the Kaluza–Klein relation for the mass dimension.*

### .3 Two types of reduction

The *downward* direction (reduction of  $T^4$ ) proceeds via two geometric operations, developed here. The lossless *upward* operation (Type III, representational translation) follows in Section 8.

#### Type I: duality decompactification (mass $\rightarrow$ time)

**Definition .3.1** (Unrolling the mass dimension). The operation

$$S_m^1 \xrightarrow{\text{covering}} \mathbb{R}_t, \quad p : \mathbb{R}_t \rightarrow S_m^1, \quad t \mapsto e^{2\pi i t/R_m},$$

unrolls the compact mass circle into the non-compact time direction via its universal cover. It is governed by  $\tilde{T} \cdot m = 1$ .

**Theorem .3.2** (Type I preserves mode content). *The unrolling  $S_m^1 \rightarrow \mathbb{R}_t$  is an embedding, not a projection. Periodic structures on  $S_m^1$  lift uniquely into  $\mathbb{R}_t$ ; the function space on  $\mathbb{R}_t$  is strictly larger. No mode content is forgotten; rather, an additional non-compact direction arises – time.*

**Proof:** (Sketch) The universal cover  $p : \mathbb{R} \rightarrow S^1$  is a local homeomorphism and surjective. An  $R_m$ -periodic function on  $\mathbb{R}$  corresponds bijectively to a function on  $S_m^1$  (Fourier series, discrete spectrum). The reverse direction (compactification  $\mathbb{R} \rightarrow S^1$ ) is lossy, since non-periodic parts do not descend. Unrolling therefore proceeds in the *embedding*, not the *projecting*, direction.

**Corollary .3.3** (Time as embedding price). *The “price” of decompactification is not loss of information but the appearance of the time direction itself (Doc. 185). What does not remain directly visible is only the compactness: the periodicity of  $S_m^1$  appears in the observable not as geometry but indirectly – as the discrete structure of the spectrum.*

**Remark .3.4** (What “not totally loss-free” refers to). This is not about absolute loss but about *reading-dependent visibility*. In the mass reading the compactness is manifest – the spectrum is discrete and the identification  $t \sim t + R_m$  is expressed directly. In the time reading the same compactness is only *latent*: from continuous time alone the topological identification is not uniquely reconstructible (the winding remains ambiguous). The duality guarantees full translatability of the spectrum; the identification is carried by the time reading only implicitly – through the discreteness of the spectrum. This is exactly where  $\xi^n$  acts.

#### What makes the unrolled time continuous and uniform

Two properties of the unrolled time have two different causes — and a third question (*that it unrolls at all*) has an entirely different answer.

**Continuity: the cover.** That  $\mathbb{R}_t$  is *continuous* is a topological fact, not an addition: the unrolling  $S_m^1 \rightarrow \mathbb{R}_t$  is the universal cover  $p : \mathbb{R} \rightarrow S^1$  (Thm. .3.2), a local homeomorphism. The cover of a circle is the continuous line. Hence, as soon as a circle is unrolled, the result is necessarily continuous — and as an embedding, not a projection (no mode content is lost).

**Uniformity: homogeneity and the fixed eigenvalue.** That time flows at a *constant rate* rests on the homogeneity of the mass circle:  $S_m^1$  is a  $U(1)$ , translation-invariant, no point distinguished. The constant, translation-invariant metric lifts to the cover  $\mathbb{R}$  as an affine, uniform parametrization. The rate itself is the *single fixed eigenvalue*  $m = 1/\tilde{T}$  (the rest energy), not a variable quantity. In the recursion picture: the constant multiplier  $\xi$  has a well-defined generator  $\ln \xi$ , so the discrete iteration  $\xi^n$  becomes the continuous dilatation flow  $t = t_0 \xi^s$  (Doc. 285); constant multiplier  $\Rightarrow$  constant rate.

**That it unrolls at all: the measurement.** Geometrically, in the compact  $T^4$  all four circles are equal — there is no time. Not the geometry but the *measurement* (the embedded observer, Doc. 248) selects one circle as time and unrolls it; a physical act, not a mathematical limit (Doc. 285). In doing so the winding number is discarded.

**The heart of it: continuity = discarded winding.** The real answer to “what makes time continuous” is therefore the discarding of the winding. The winding is the discrete, compact information; in the time reading it is thrown away (the information-discarding step, Doc. 285), and the compactness becomes *latent* — from continuous time alone the winding is no longer uniquely reconstructible (Rem. .3.4). Continuous time is thus the circle with its winding forgotten; keep the winding and the discrete circle returns.

**A subtlety: continuous exactly, uniform only up to  $O(\xi)$ .** Two properties must be separated here. *Continuity* stays exact:  $g_{\mu\nu} = \eta_{\mu\nu}(1 + \xi f)$  ( $D_f = 3 - \xi$ ) is a *metric* correction, not a topological intervention — a bounded  $\xi$ -small modulation does not tear  $\mathbb{R}$ , no gaps arise. *Uniformity*, by contrast, holds only up to  $O(\xi)$ : the same  $\xi$  feedback that in the dynamic sector carries the AM sidebands (Doc. 286) makes the rate wobble by relative  $\xi \approx 10^{-4}$ ; the flat metric  $\eta$  “ignores the fractal structure” (Doc. 051). The perfectly smooth, uniform time is thus the limit  $\xi \rightarrow 0$  — the idealized time of relativity —, while the real FFGFT time is a continuum *with* a  $\xi$ -fine fractal rate modulation. Important: this is *not* discrete — fractal  $\neq$  discrete; time remains an uncountable continuum, only not smoothly affine. (Whether this  $\xi$  graininess is the metric shadow of the latent compactness — a  $\xi$  remnant of the discarded winding — is a natural conjecture but not established: the latent compactness shows in the discrete spectrum (Rem. .3.4), the  $\xi$  correction in the metric.)

(Note of 2 Oct. 2026 on notation: here  $f$  in  $g = \eta(1 + \xi f)$  is the *fractal correction function* of the metric, not a frequency and not the base winding number  $f = 1/\xi$ . Cf. Doc. 190, R147.)

## Type II: geometric projection (spatial)

**Definition .3.5** (Forgetting a spatial direction). A projection  $\pi : \mathbb{R}^k \rightarrow \mathbb{R}^{k-1}$  with  $\ker(\pi) = \mathbb{R}$  forgets a spatial direction. It is not injective and therefore not invertible.

**Theorem .3.6** (Type II is strictly lossy). *In a spatial projection  $\mathbb{R}^k \rightarrow \mathbb{R}^{k-1}$  a full fibre  $\mathbb{R}$  collapses. The information about the forgotten direction is not contained in the image and cannot be reconstructed without additional data (boundary conditions, holographic completeness). The recursion  $\xi^n$  does not act on this step.*

|                 | Type I (mass→time)        | Type II (spatial)       |
|-----------------|---------------------------|-------------------------|
| Operation       | unrolling / covering      | forgetting / projection |
| Direction       | embedding                 | projecting              |
| Mode content    | preserved                 | lost                    |
| Reversibility   | dual/spectral recoverable | not reversible          |
| Role of $\xi^n$ | spectral signature        | ineffective             |
| Residue         | compactness only inferred | genuine geometric loss  |

## .4 The formal chain, step by step

### Step 1: $D4 \rightarrow D3 + \text{time}$ (Type I)

$$T^4 \longrightarrow \underbrace{T^3}_{\rightarrow \mathbb{R}^3} \times \underbrace{\mathbb{R}_t}_{\text{from } S_m^1}.$$

Step 1 bundles two information-preserving operations of *different* mechanism: (Ia) the *mass circle* unrolls to  $\mathbb{R}_t$  via the duality  $\tilde{T}m = 1$  per Theorem .3.2 (duality decompactification); (Ib) the three *spatial circles* decompactify through the *limiting process* of large radii ( $R_{x,y,z} \rightarrow \infty$  for  $r \ll R_H$ ) to  $\mathbb{R}^3$  (limit decompactification). Both are embeddings, not projections; but (Ib) is *not* a duality step, it is a limit – see the clarification in Section .12, point 3.

*Invariant:*  $\tilde{T} \cdot m = 1$  (Prop. .2.4); in the spatial limit, local flatness. *Loss:* none in mode content; only the visibility of compactness (Rem. .3.4).

### Step 2: $D3 \rightarrow D2$ (Type II, holographic)

$$\pi_2 : \mathbb{R}^3 \longrightarrow \mathbb{R}^2, \quad \ker(\pi_2) = \mathbb{R} \text{ (depth direction).}$$

*Invariant:* area  $A$ . *Holography:* step 2 is mathematically the holographic principle; in the horizon case the Bekenstein–Hawking bound  $S \leq A/(4\ell_P^2)$  (cf. Doc. 254/263). *Loss:* genuine depth direction.

### Step 3: $D2 \rightarrow D1$ (Type II)

$$\pi_3 : \mathbb{R}^2 \longrightarrow \mathbb{R}, \quad \ker(\pi_3) = \mathbb{R}.$$

*Invariant:* length  $L$ . *Loss:* second spatial direction; no area measure remains.

### Step 4: $D1 \rightarrow D0$ (Type II, total loss)

$$\pi_4 : \mathbb{R} \longrightarrow \{\text{pt}\}.$$

*Invariant:* only a discrete scalar measure ( $n \in \mathbb{Z}$ ). *Loss:* the complete geometric structure.

## .5 Reversibility, precise per type

**Theorem .5.1** (Reversibility of the full chain). *Let  $\Pi = \pi_4 \circ \pi_3 \circ \pi_2$  be the spatial subchain  $\mathbb{R}^3 \rightarrow \{\text{pt}\}$  and  $D$  the Type I operation  $S_m^1 \rightarrow \mathbb{R}_t$ . Then:*

- (i)  *$D$  is, as a duality (Fourier/KK), invertible in mode content; the winding remains ambiguous.  $D$  is “almost reversible”.*
- (ii)  *$\Pi$  is not invertible: there is no continuous map  $\rho : \{\text{pt}\} \rightarrow \mathbb{R}^3$  with  $\rho \circ \Pi = \text{id}$ .*

**Proof:** (Sketch) (i) follows from Theorem .3.2: unrolling is embedding, the discrete spectrum encodes the full periodic content; only the choice of lift (winding) is non-unique. (ii) follows from Theorem .3.6: each  $\pi_k$  has  $\ker = \mathbb{R} \neq \{0\}$ . A left inverse would have to reconstruct the collapsed fibre structure; this information is not present in the image  $\{\text{pt}\}$ .

**Corollary .5.2** (Resolution of the initial tension). *Pictures (A) and (B) are both correct but concern different operations. The non-reversibility (A) holds for the spatial projections (Type II). The decompactification to the time direction (B) is the embedding Type I step – it creates a dimension instead of forgetting one. There is no contradiction.*

## .6 Role of the recursion $\xi^n$

The recursion  $R : \xi^n \mapsto \xi^{n+1}$  is *not* a universal compensation for projection losses. Its function is narrowly bounded:

- **Type I:**  $\xi^n$  generates the discrete mass spectrum (particle ladder). This spectrum is the compact mode content of  $S_m^1$ , dually expressed as time-frequency structure.  $\xi^n$  “fills in” nothing; it is the spectral signature through which the hidden compactness remains recognisable (Rem. .3.4).
- **Type II:**  $\xi^n$  has no effect. Spatial projection losses are genuine and untouched by the recursion.

*Remark .6.1* (Correction of a tempting misrepresentation). A presentation that assigns the recursion a “compensation strength” decreasing monotonically across all four steps (full/partial/weak/none) is misleading. It presupposes a uniform lossy ladder that does not exist. The recursion belongs exclusively to the Type I step.

## .7 Overview

| Step                  | Operation                                 | Type | Invariant          | $\xi^n$            |
|-----------------------|-------------------------------------------|------|--------------------|--------------------|
| $D4 \rightarrow D3+t$ | $S_m^1 \rightarrow \mathbb{R}_t$ (unroll) | I    | $\tilde{T}m = 1$   | spectral signature |
| $D3 \rightarrow D2$   | forget space                              | II   | area $A$           | –                  |
| $D2 \rightarrow D1$   | forget space                              | II   | length $L$         | –                  |
| $D1 \rightarrow D0$   | forget space                              | II   | $n \in \mathbb{Z}$ | –                  |

**Table 1:** The chain  $T^4 \rightarrow T^0$  separated by type.

$$\underbrace{T^4 \longrightarrow \mathbb{R}^3 \times \mathbb{R}_t}_{\text{Type I: embedding, almost reversible}} \longrightarrow \underbrace{\mathbb{R}^2 \longrightarrow \mathbb{R} \longrightarrow \{\text{pt}\}}_{\text{Type II: projecting, not reversible}}$$

## .8 The upward direction: representational translation (Type III)

The chain so far describes only the *downward* direction (reduction). There is a counter-direction – but it is *not* geometric.

**Definition .8.1** (Type III: representational translation). The operation

$$T^4 \xleftrightarrow{\text{bijective}} \mathcal{H}$$

translates the FFGFT mode formalism on  $T^4$  into the (infinite-dimensional) Hilbert space  $\mathcal{H}$ , or the equivalent matrix algebra (Doc. 230, 206, 231). It is not an approximation but a *bijective structural statement*: every QM computation is expressible in FFGFT language and vice versa.

### The dimensional asymmetry

Downward (geometric) and upward (representational) are different kinds of operation with *opposite* information properties:

- **Downward (Type II, geometric)**: lossy, not reversible – a fibre collapses (Theorem .3.6).
- **Upward (Type III, representational)**: bijective, lossless – it adds *no* physical dimensions but expresses the same 4D content in a higher-dimensional *language* in which it fits without reduction.

The losslessness follows *precisely from* the representational character – not from any gain of genuine extra dimensions.

*Remark .8.2* (No geometrically higher dimensions). A geometric extension  $T^4 \rightarrow T^5 \rightarrow \dots$  is excluded in FFGFT (Doc. 171): complexity arises from *fractal depth* ( $D_f = 3 - \xi$ , self-similarity), not from *more* dimensions. This is the deliberate contrast with string theory (10/11D, Calabi-Yau, landscape problem). “Higher” therefore always means Type III (representational) in FFGFT, never geometric.

FFGFT thus has three dimension-changing operations:

| Type | Operation                                        | Character        | Information       |
|------|--------------------------------------------------|------------------|-------------------|
| I    | mass $\leftrightarrow$ time ( $\tilde{T}m = 1$ ) | geometric        | almost reversible |
| II   | spatial projection (down)                        | geometric        | lossy             |
| III  | $T^4 \leftrightarrow \mathcal{H}$ /matrix (up)   | representational | <b>bijective</b>  |

The bijectivity of Type III is the mirror image of the non-reversibility of Type II (Theorem .5.1): geometrically one loses on reducing, representationally one loses nothing on translating.

## .9 Why exactly four circles? Minimal sufficiency, units, and visualization

The chain so far (Type I/II downward, Type III upward) presupposes the four-circle structure  $T^4 = S_{x,y,z}^1 \times S_m^1$ . This section answers *why* there are exactly four – deliberately *not* as a derivation of the number, but as *minimal sufficiency*.

### 3 + 1 is input, not derivation

No theory derives the dimension count of spacetime: the Standard Model and GR assume  $3 + 1$ ; string theory fails on uniqueness (landscape problem). FFGFT, too, *does not derive the 4 forward*; it takes  $3 + 1$  as *empirical input*. The question is therefore not “why is the world four-dimensional?” but “*given*  $3 + 1$  – which structure is the minimal one that carries everything?”.

## Minimal sufficiency: four instead of five

### The duality saves one dimension

Naively one would need *five* building blocks: three for space, one for a mass spectrum (particles), one for time evolution. The fundamental relation  $\tilde{T} \cdot m = 1$ , however, identifies the *mass reading* and the *time reading* of *one* dimension (Type I, Theorem .3.2): the fourth circle  $S_m^1$  carries the mass spectrum as its KK tower *and*, decompactified, time. The duality lets *one* circle do the work of two – so the naive 5 collapses to **4**.

- **Fewer** (only  $T^3$ ): no mass spectrum, no time.
- **Four** ( $T^3$  + one dual circle): space, masses *and* time from *one* object, without redundancy.
- **More** (a second such circle): a second independent mass tower or a second time – superfluous, without correlate.

“Optimal” therefore means *minimal-sufficient*: the smallest dimension count in which space, the full mass spectrum, and time jointly emerge from one geometric object.

### The four circles as units bookkeeping

The four-circle structure *is* at the same time the exponent bookkeeping of the SI conversion factors – each circle type carries its bridge:

$$\underbrace{S_{x,y,z}^1}_{\text{space}} \xrightarrow{c} \text{time}, \quad \underbrace{S_m^1}_{\text{mass}} \xrightarrow{\hbar (\tilde{T}m=1)} \text{time}.$$

$c$  is the space-time bridge of the three spatial circles,  $\hbar$  the mass-time bridge of the mass circle. This is why the SI conversions are *indispensable* once one wants to determine the exponent structure correctly:

**Remark .9.1** (Why natural units hide the powers). Natural units ( $c = \hbar = 1$ ) collapse  $L$ ,  $T$  and  $M$  to *one* axis. This removes precisely the distinction that carries the exponent structure – the independent *power of  $c$*  resp. *of time*.  $c$  and  $\hbar$  may be set to 1 only *after* the exponents have been read off; before that the information is destroyed. The factors are the scaffolding that keeps the powers visible – and easy to handle precisely because their whole role is that scaffolding without a free parameter.  $G$ , too, is *derived* – reconstructed from  $\ell_P$ ,  $c$ ,  $\hbar$ , for which it needs only the SI relations (Docs. 012/056) – and carries *no* free factor. The only free factor sits not in a conversion or natural constant but in the *cosmological absolute scale*  $R_H$  (exponent 41/4; Doc. 190, R20).

### The same structure as Hilbert visualization

The four-circle geometry is also the *faithful* visualization of the infinite-dimensional Hilbert space (Type III, Def. .8.1) – two readings of *the same* bijection:

- **generative/ontological**:  $T^4 \rightarrow \mathcal{H}$  – the geometry *generates* the algebra (Doc. 206), not the reverse;
- **visualizing/epistemic**: the abstract  $\mathcal{H}$  becomes intuitable at all only through the 4D geometry.

“Higher-dimensional” (the  $\infty$ -dimensional  $\mathcal{H}$ ) does *not* mean “more fundamental”: the generative direction is  $T^4 \rightarrow \mathcal{H}$ . Precisely *because* the 4D structure is generative (not a mere picture), the visualization is *exact* – bijective, not a cartoon.

## Summary of the dimension line

### What "optimal" means

$T^4$  resp.  $T^3 + 1$  time is the *minimal-sufficient, dual-economical and Hilbert-faithful* configuration – *given*  $3 + 1$ . The strength lies in the economy (one dimension saved by  $\tilde{T}m = 1$ ) and the faithfulness (lossless Hilbert representation), *not* in a uniqueness of the number. The bare 4 is input, not a forward result (Doc. 190, R34) – the same signature as for the  $\xi$ -magnitude (R33) and the cosmological scale (R20): *structure forward, number input*.

## .10 In the compact view, time is just another circle: the asymmetry is the measurement

In the compact representation  $T^4 = S^1 \times S^1 \times S^1 \times S^1$  all four circles are identical: dimensionless cyclic coordinates, no signature, no space/time distinction. Time is written *exactly* as the three spatial circles; at the compact level the Lorentzian asymmetry does not exist. This is why, in FFGFT, the time coordinate is simple to write – it is one circle like the others. The complication enters only when one leaves the compact picture.

The asymmetry (one time, signature  $(3, 1)$ ) appears only under decompactification (Type I, Def. .3.1): exactly one circle – the mass circle,  $\tilde{T} \cdot m = 1$  – unrolls through its universal cover into non-compact time, while the other three stay compact (discrete spectrum) or project (space, Type II).

*Which* circle becomes time is therefore not an intrinsic-geometric fact to be derived. The geometry is symmetric, and nothing in it should single one direction out. Time is *relational*: the selection is the measurement itself. The decompactification *is* the measurement – the read-out/projection of this document and of Doc. 284 – and committing one circle to non-compact, experienced time is exactly the information-discarding step a measurement performs. The embedded observer, reading out from the inside (Doc. 248, self-referentiality), experiences one direction as duration; that experience constitutes the time circle. The symmetry-breaking is the measurement, not the geometry.

[The time circle is selected by the measurement] In the compact view the four circles of  $T^4$  are symmetric. Time is not chosen by the geometry but is relational: the symmetry-breaking *is* the measurement (the decompactification of Type I). Three statements then close the bookkeeping with no gap:

- **That there is one time** (not two) is the signature/predictivity constraint – one timelike direction gives hyperbolic field equations with a well-posed Cauchy problem; two or more give ultrahyperbolic equations with no well-posed initial-value problem.
- **Which circle is time** is fixed by the measurement, i.e. by the experienced situation of the embedded observer (Doc. 248).
- **That the split is  $3+1$**  is the structure of being an embedded observer – the empirical input of Doc. 190 (P34), read here as the observer's situation rather than a free parameter.

The contrast with a non-compactified treatment of time is then sharp. A framework that writes time as a complicated structure built on already-physical time  $t$  (for instance a spiral-time operator  $\psi(t) = t + i\varphi(t) + j\chi(t)$ , Doc. 283/284) has *reified* the result of the measurement into its kinematics, instead of recognising experienced time as the measurement-induced

decompactification of a symmetric circle. In the compact view that complication is unnecessary: time is one more circle, and only the read-out turns it into the non-periodic axis we experience.

## .11 Physical testability of the identification $t \sim t + R_m$

The topological identification  $t \sim t + R_m$  is not a mere accompanying structure of the duality but a physical observable. In the time reading its period is *exactly* the fundamental revolution time

$$T_{\text{rev}} = \frac{R_m}{c} = \frac{\xi \ell_P}{c} \approx 7.2 \times 10^{-48} \text{ s}, \quad \Delta\phi = 2\pi\xi \approx 8.4 \times 10^{-4} \text{ rad/revolution},$$

so that  $t \sim t + R_m$  is identical with  $t \sim t + cT_{\text{rev}}$  (Doc. 183/185). The compactness of the fourth direction *is* the revival time.

### Two testable readings

- **Mass reading (mature, quantitative).** The discrete mode spectrum generates the particle masses as torus overtones with step  $1/D$ ; from this follow Koide,  $g-2$  and the three generations (Doc. 159, 258/259). Here the compactness is numerically realised.
- **Time reading (falsifiable in principle, not resolvable with present hardware).** The periodicity produces a CHSH deviation from the Tsirelson bound of order  $\xi$ . For 73 qubits FFGFT predicts  $\Delta \approx 5.4 \times 10^{-4}$ . The hardware test IBM Heron r2 (May 2026) confirms the Bell violation ( $S = 2.74$ ) and the executability of T0 circuits; the  $\xi$  effect, however, lies far below the hardware noise and is not resolvable with present hardware (Doc. 147). *(Corrected on 1 Oct. 2026: previously "currently below resolution", "resolvable at  $\sigma \sim 10^{-4}$ " and "not yet resolvable" –  $\sigma \sim 10^{-4}$  is only the statistical bound; the measured deviation from the Tsirelson bound is  $2\sqrt{2} - 2.74 = 0.088$  (3.1%), about 160 times  $\Delta$ ; by Doc. 190 R142/R143 the  $\xi$  effect is not resolvable with present hardware.)*

### Scaling: linear versus logarithmic

#### Regime dependence of phase accumulation

A naive *coherent linear accumulation*  $r \cdot \Delta\phi$  over  $r$  revolutions would yield physically absurd phase drifts (of order  $10^{44}$  rad/s) and contradict working atomic clocks and deep quantum circuits. The lab-testable prediction therefore scales *logarithmically*,

$$\delta\phi \sim \frac{\xi \ln(r)}{D_f} \approx 2 \times 10^{-4} \quad (D_f = 3 - \xi),$$

not linearly. The linear form  $r \cdot \Delta\phi$  of Doc. 185 is the *worst-case bound of the RSA argument* (coherence over  $r \approx N$  cycles), not the lab prediction. This regime separation explains why deep coherent circuits can run while the RSA factorisation of large keys remains geometrically protected.

**Conclusion.**  $t \sim t + R_m$  is carried in the FFGFT corpus as an observable ( $T_{\text{rev}}$ , CHSH deviation  $\sim \xi$ ): quantitatively confirmed in the mass reading, falsifiable in principle in the time reading, but not resolvable with present hardware (far below the hardware noise). *(Corrected on 1 Oct. 2026: previously "currently just below experimental resolution" –*

see the note on the time reading above; Doc. 190 R142/R143.) It is not an unfalsifiable accompanying structure.

## .12 Classification and open questions

**Connections.** Doc. 185 (Time as the Embedding Price) provides the ontological basis of the Type I step; Doc. 263 (holographic projection) and Doc. 254 (area-bound theorem) the framework for step 2; Doc. 206 (Triangle–Matrix reduction) confirms that the 4D level is generative; Doc. 147/183 provide the time-side testability. The upward operation (Type III) is developed in Doc. 230 (bijective Hilbert-space translation), Doc. 231 (necessary extensions) and Doc. 206 (matrix representation); Doc. 171 justifies the exclusion of geometrically higher dimensions.

### Open points.

1. **Structural assumption – confirmed.** The decomposition  $T^4 = S_{x,y,z}^1 \times S_m^1$  with  $S_m^1$  as substitute-time direction and  $\tilde{T} \cdot m = 1$  as translation is fixed (Rem. .2.2).
2. **Linear/logarithmic scaling.** The linear worst-case bound  $r \cdot \Delta\phi$  (Doc. 185, RSA regime) and the logarithmic lab prediction  $\xi \ln(r)/D_f$  (Doc. 147) should be marked explicitly as regime-dependent to secure consistency of presentation.
3. **Spatial limit – clarified.** The decompactification  $S_{x,y,z}^1 \rightarrow \mathbb{R}^3$  is a *limiting process* ( $R_{x,y,z} \rightarrow \infty$ ), neither the Type I duality nor a Type II projection, but a distinct *limit-type* information-preserving operation (Ib). It is therefore carried in Step 1 explicitly apart from the mass-circle duality (Ia); both share the embedding character, not the mechanism.
4. **Holographic completeness.** To what extent can the Type II loss at  $D3 \rightarrow D2$  be reconstructed exactly (not just partially) from boundary data? Connection to Doc. 254/263.